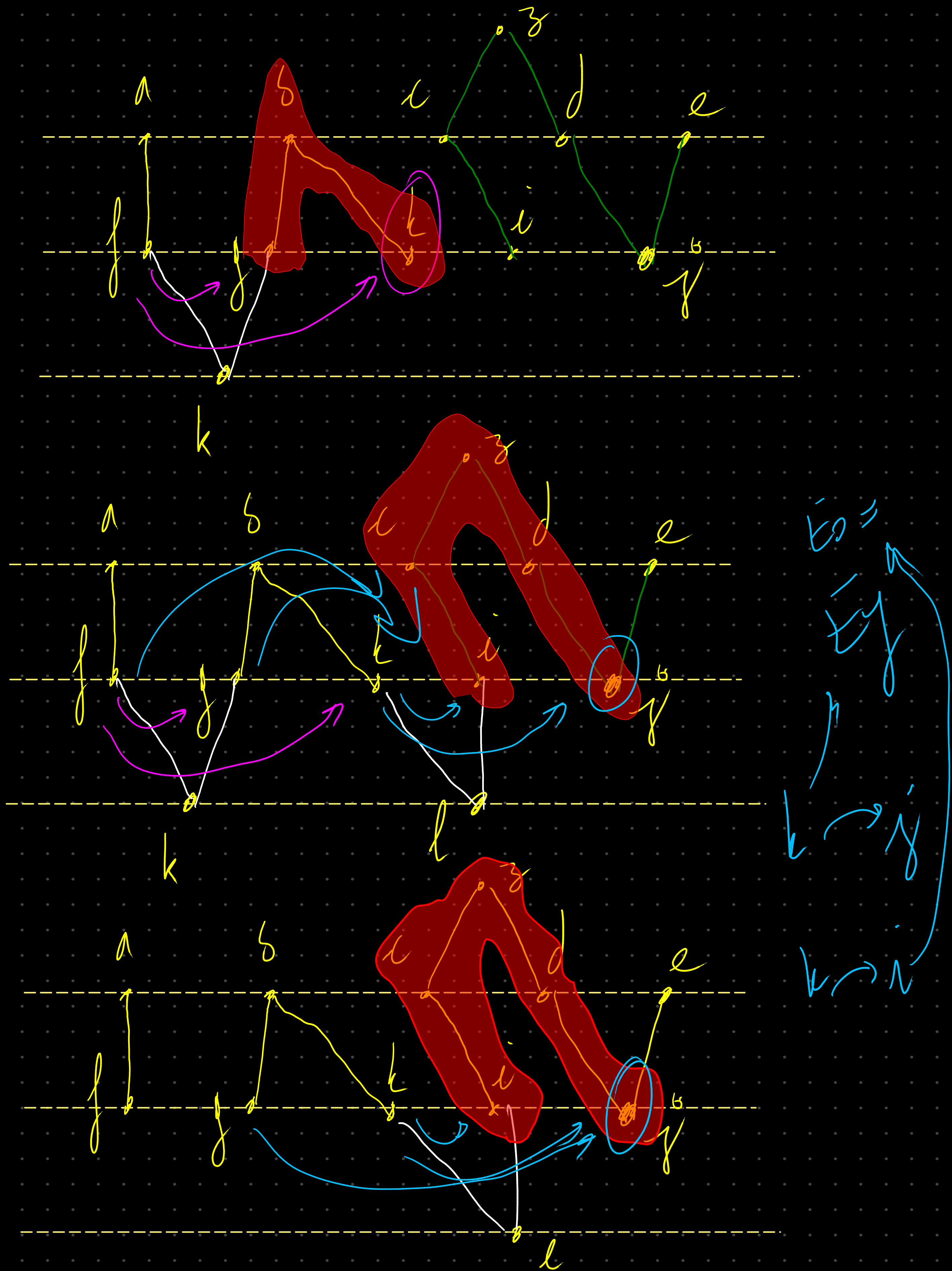
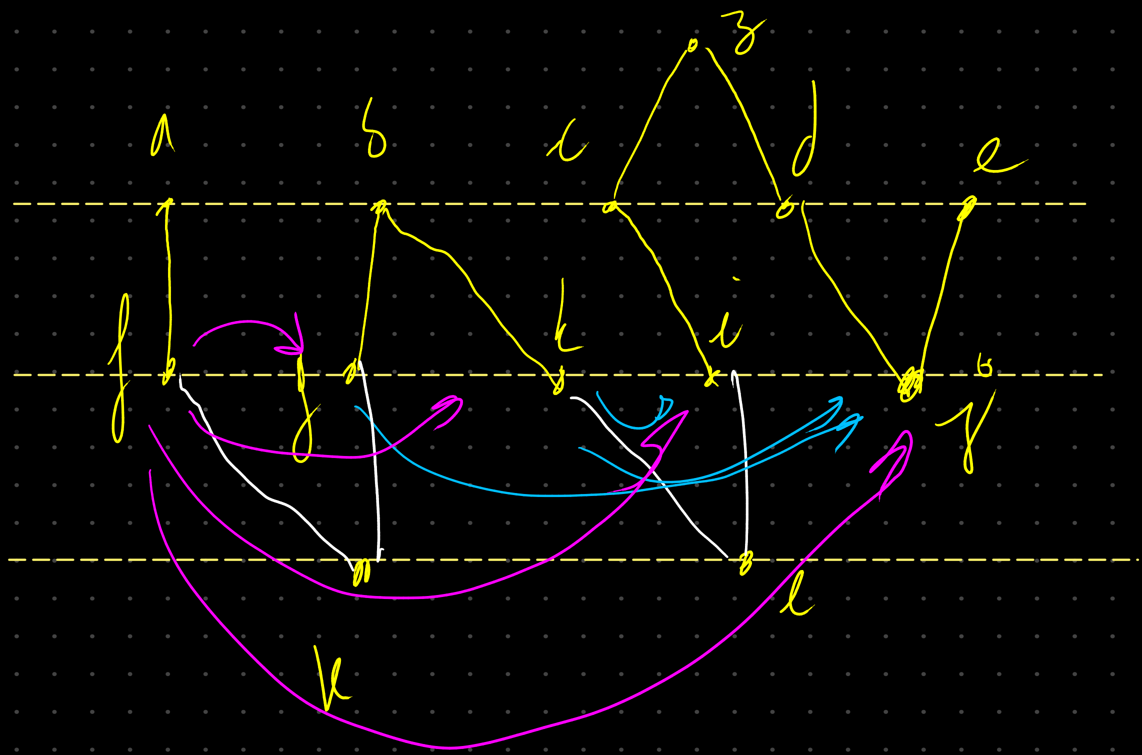
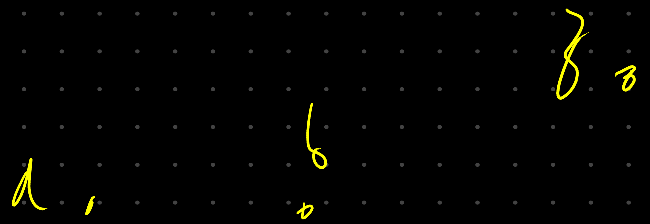




I think we need to keep track of the connected comps, while building drawing.



$C_1 = \{z\}$

$C_2 = \{a\}$

$C_3 = \{b\}$

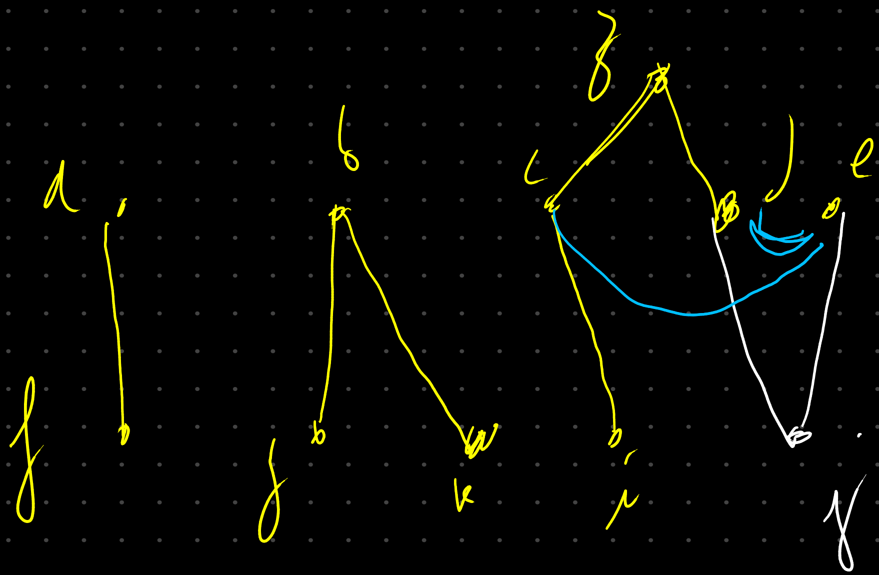
$C_1 = \{z, c, d, i\}$

$C_2 = \{a, f\}$

$C_3 = \{b, g, k\}$

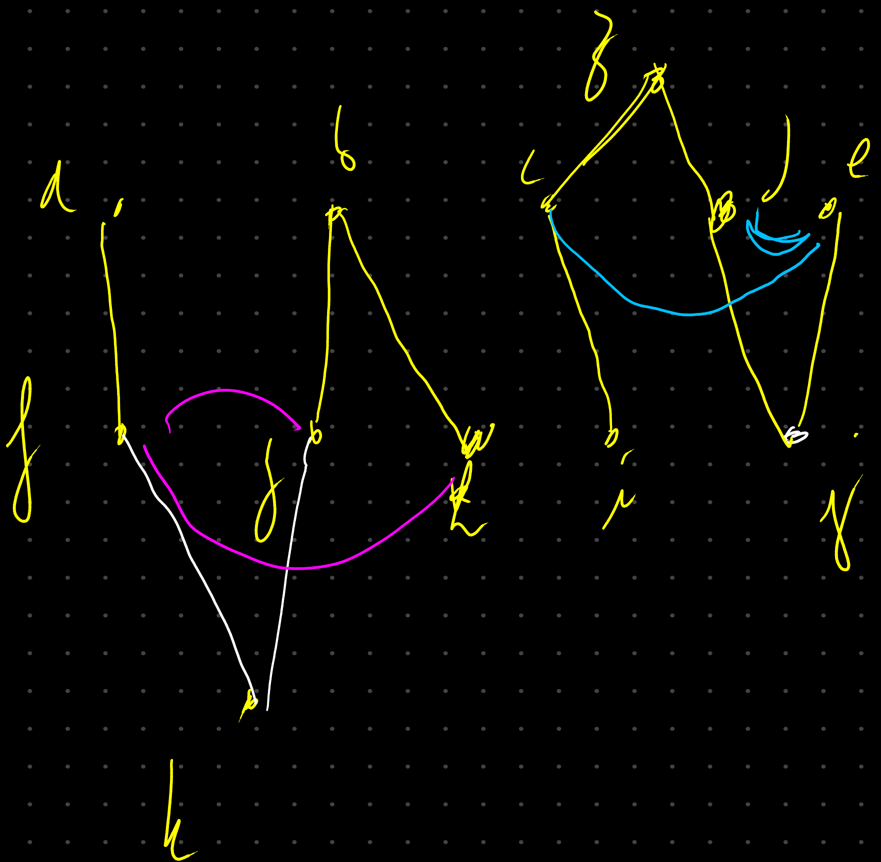
$C_4 = \{e\}$

first common endpoint of adjacent edges

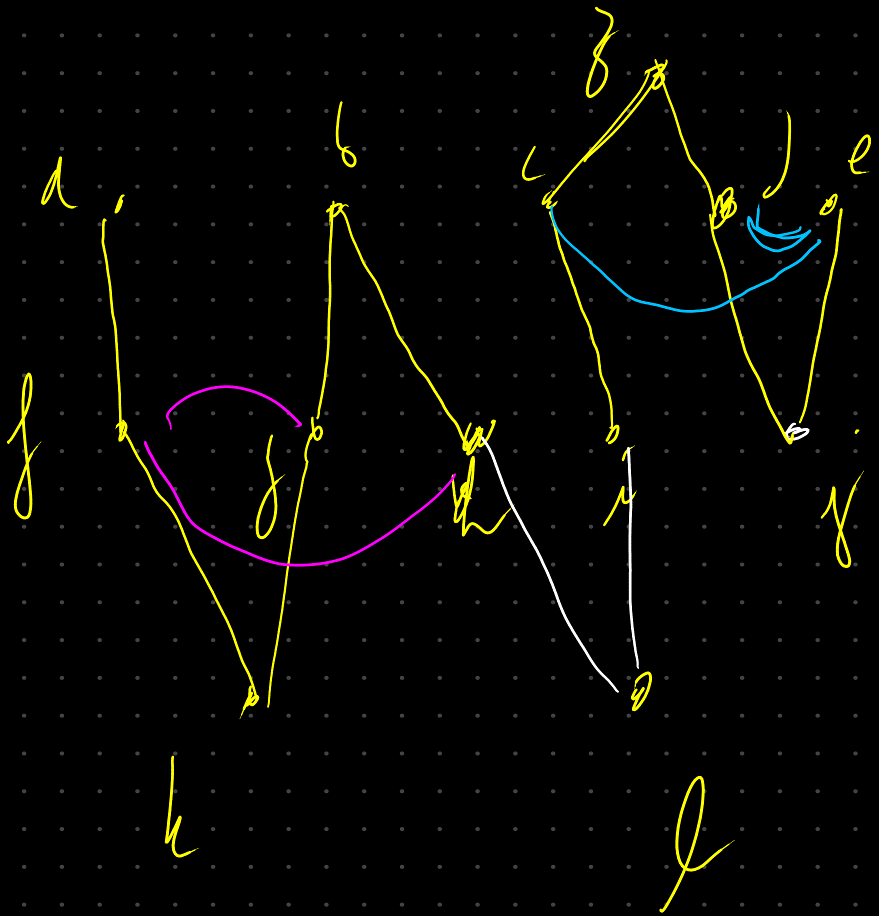


$C_1 = \{g, c, d, i\}$
$C_2 = \{a, f\}$
$C_3 = \{b, g, k\}$
$C_4 = \{e\}$

$\varphi(C_1, C_4)$

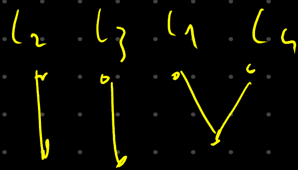


$C_1 = \{g, c, d, i, e, j\}$
$C_2 = \{a, f\}$
$C_3 = \{b, g, k\}$



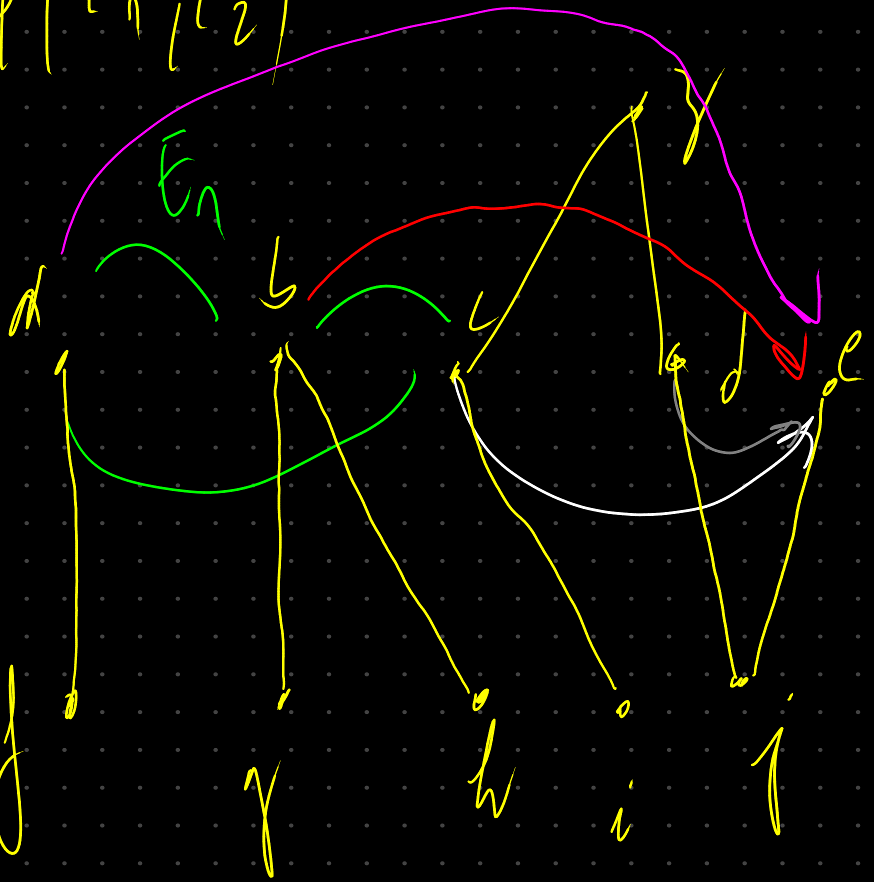
$L_1 = \{c, d, h, i, e, j\}$

$L_2 = \{a, f, g, b, k\}$



$\varphi(L_2, L_3) = \{\bar{E}_1\}$

$\varphi(L_1, L_2)$



$\varphi(L_2, L_1) = \{\bar{E}_1\}$

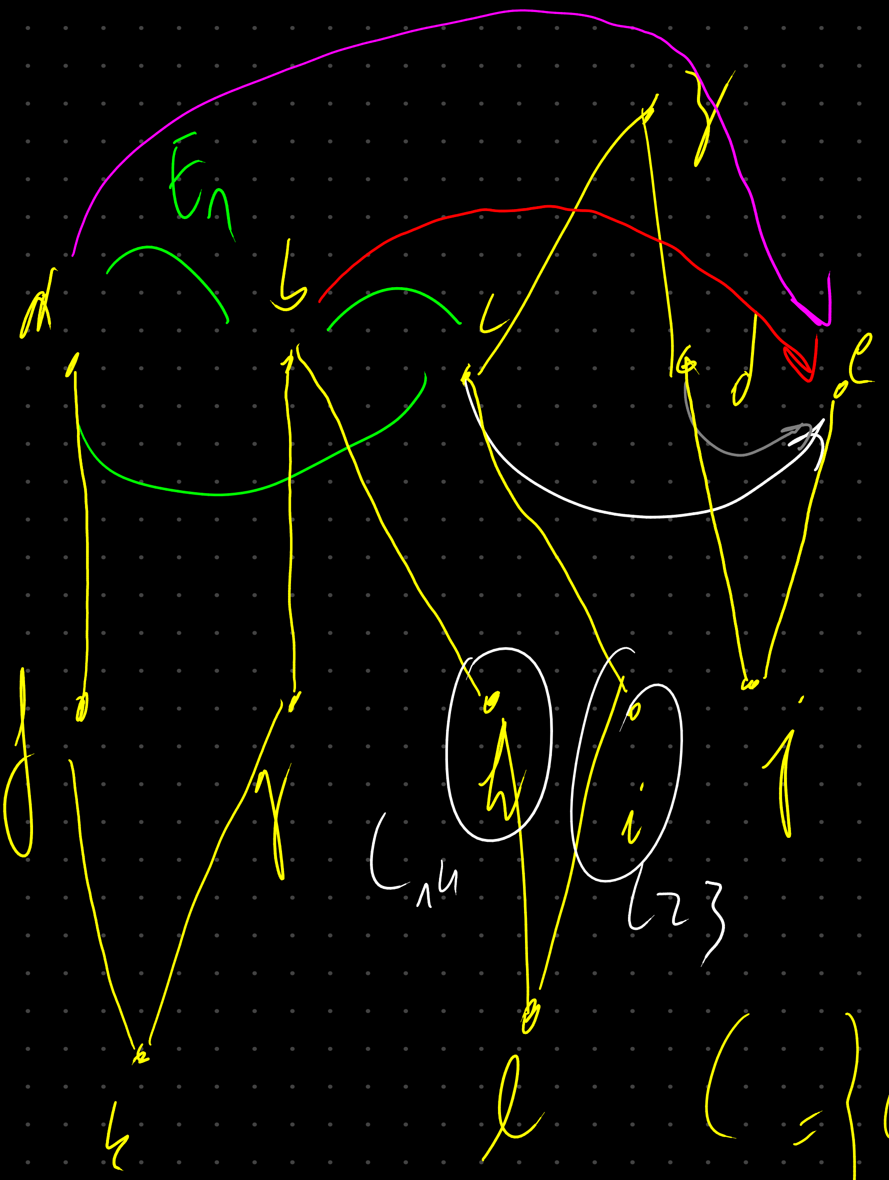
$\varphi(L_3, L_1) = \{\bar{E}_1\}$

$\varphi(L_2, L_4) = \{E_2\}$

$\varphi(L_3, L_4) = \{E_3\}$

$\boxed{\varphi(L_1, L_4)} = \{\bar{E}_4, E_5\}$

fix



$\varphi(C_2, C_1) = \{E_1\}$

$\varphi(C_3, C_1) = \{E_1\}$

$\varphi(C_2, C_4) = \{E_2\}$

$\varphi(C_3, C_4) = \{E_3\}$

$\boxed{\varphi(C_1, C_4)} = \{E_4, E_5\}$

fix

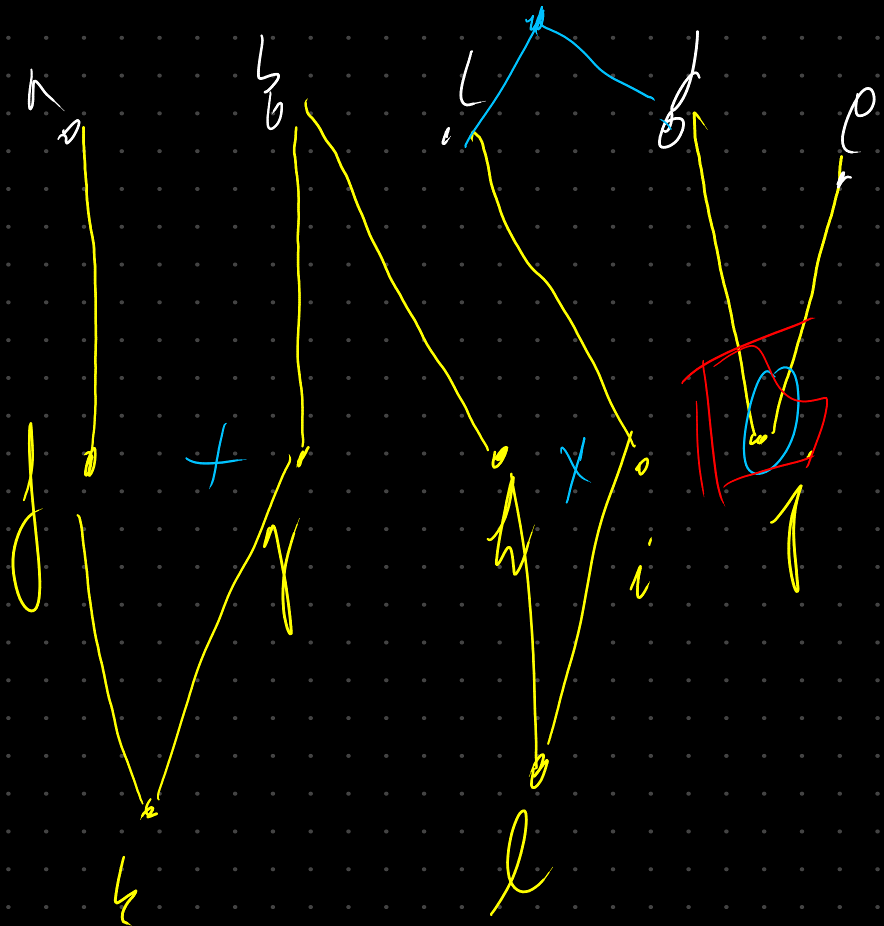
$C = \{C_{23}, C_{14}\}$

$\varphi(C_1, C_4)$ fixed

$\varphi(C_2, C_3)$ fixed

$\underbrace{E_4 = E_5}_{E_1}$

$\varphi(C_2, C_{14}) = \varphi(C_2, C_1) \cup \varphi(C_2, C_4)$

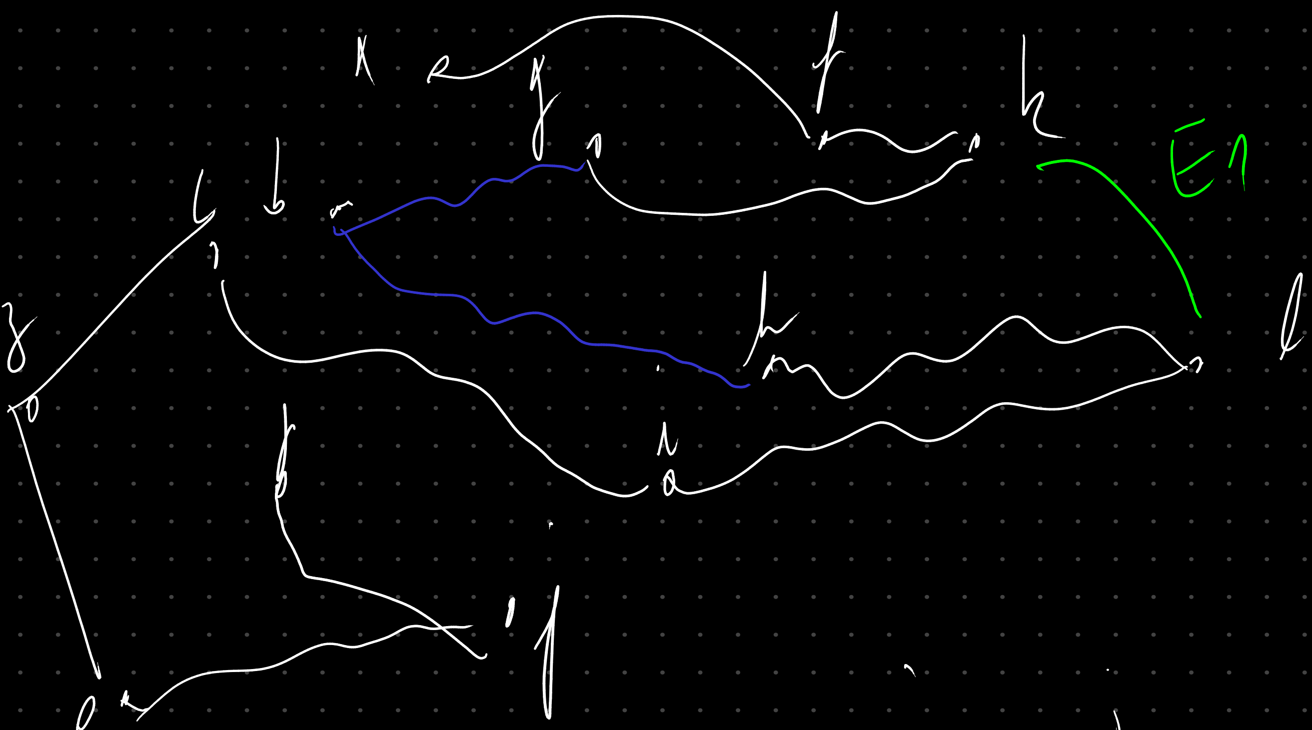


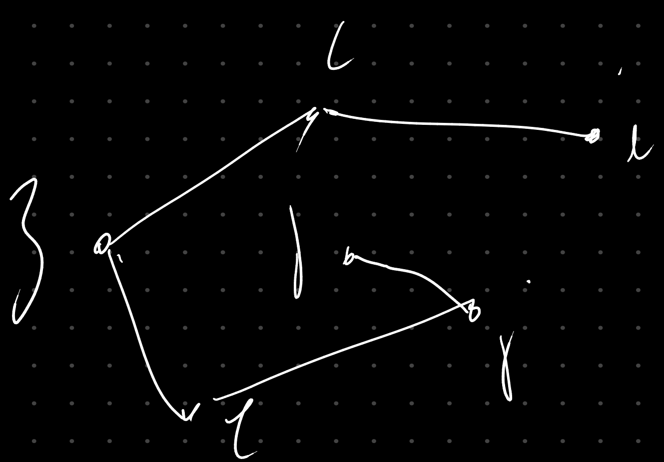
$C_1 = \{k\}$

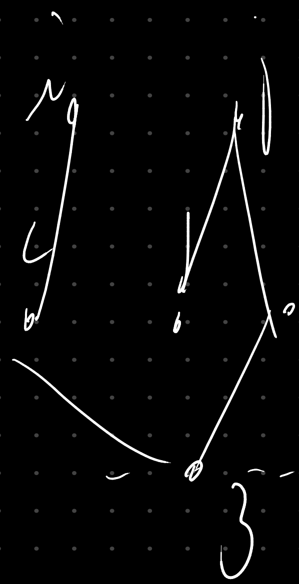
$C_2 = \{l\}$

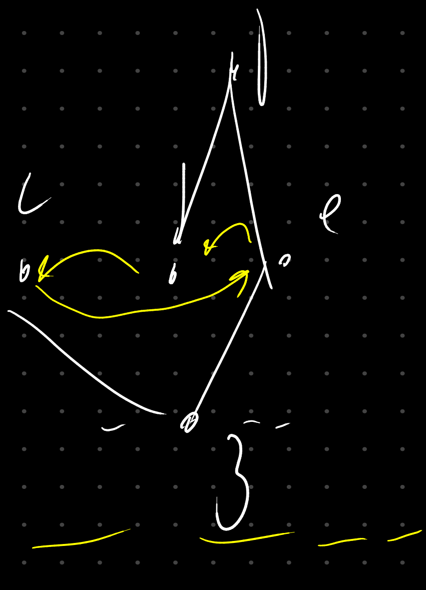
$\varphi(C_1, C_2) = \{E_1\}$

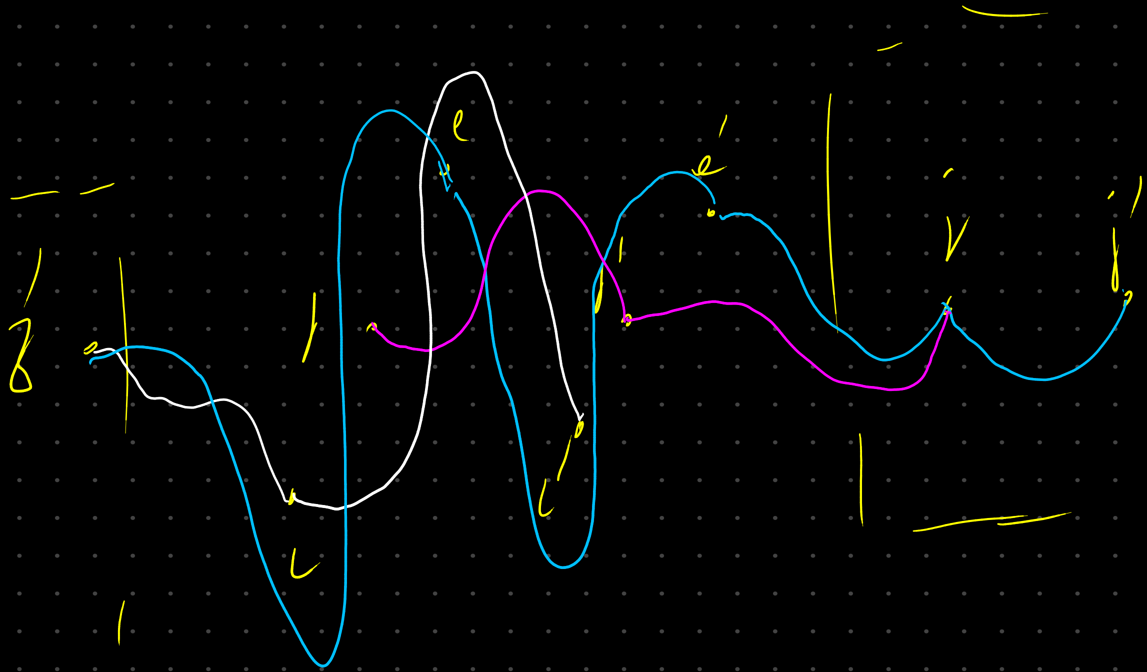
$C_1 = \{k, l$

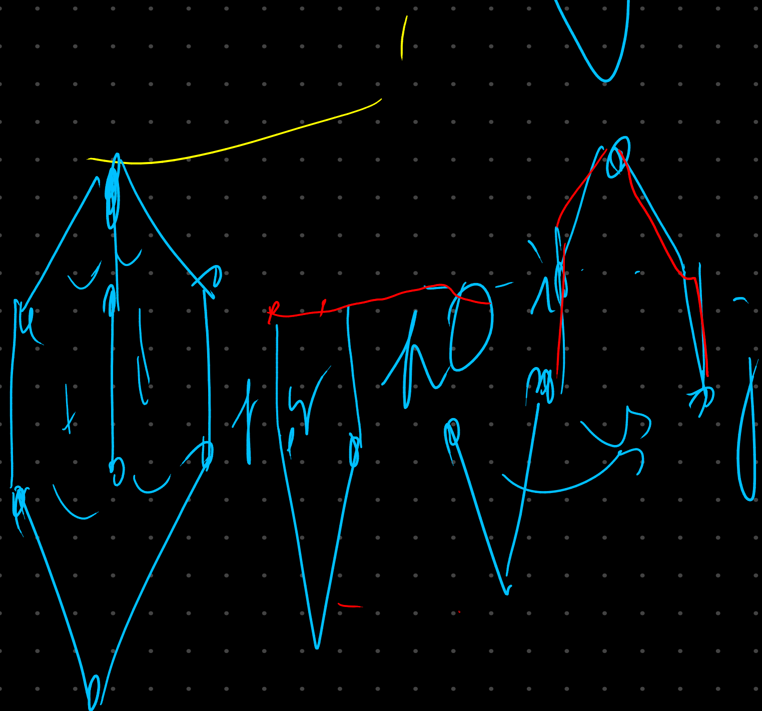


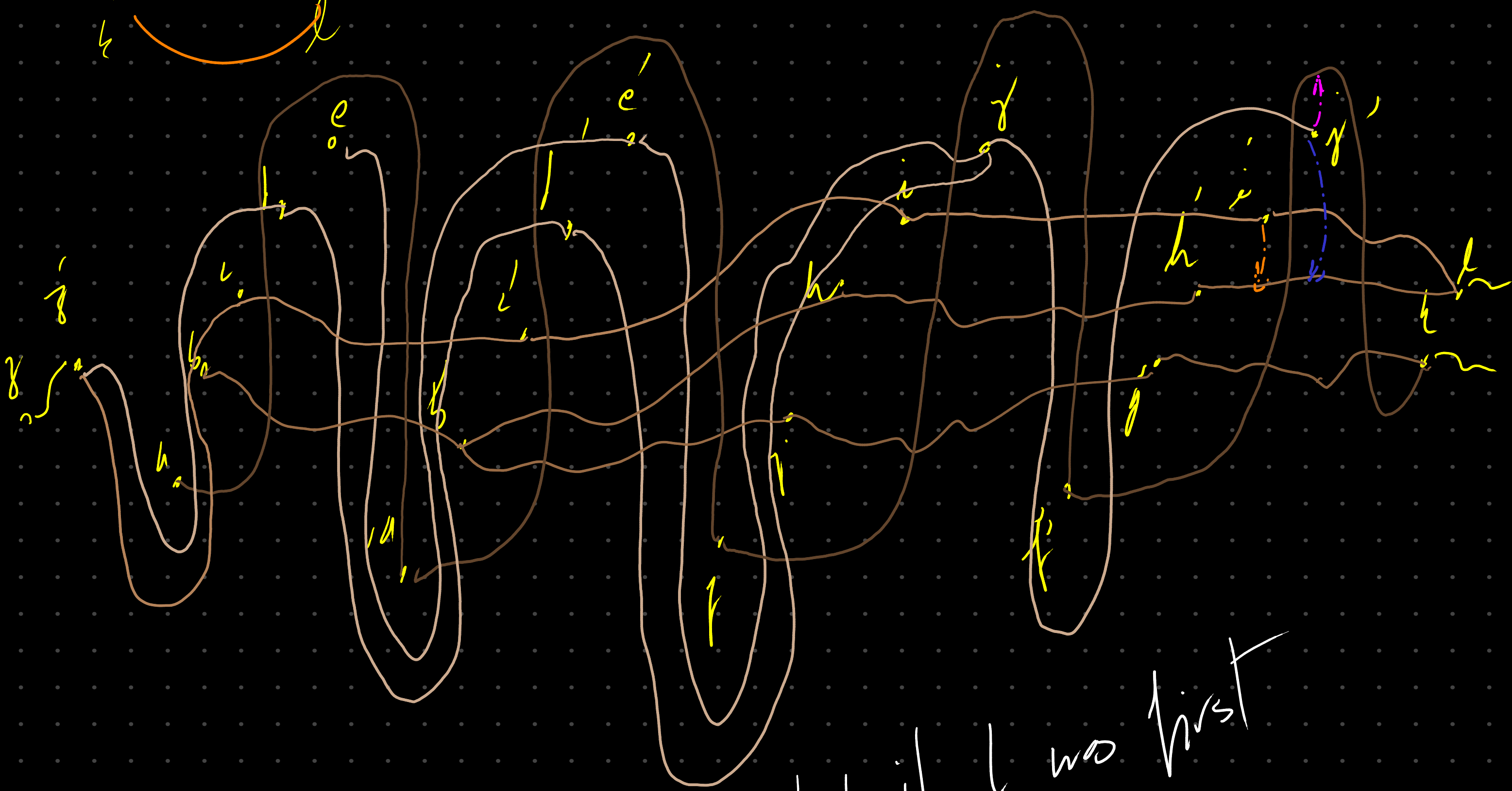






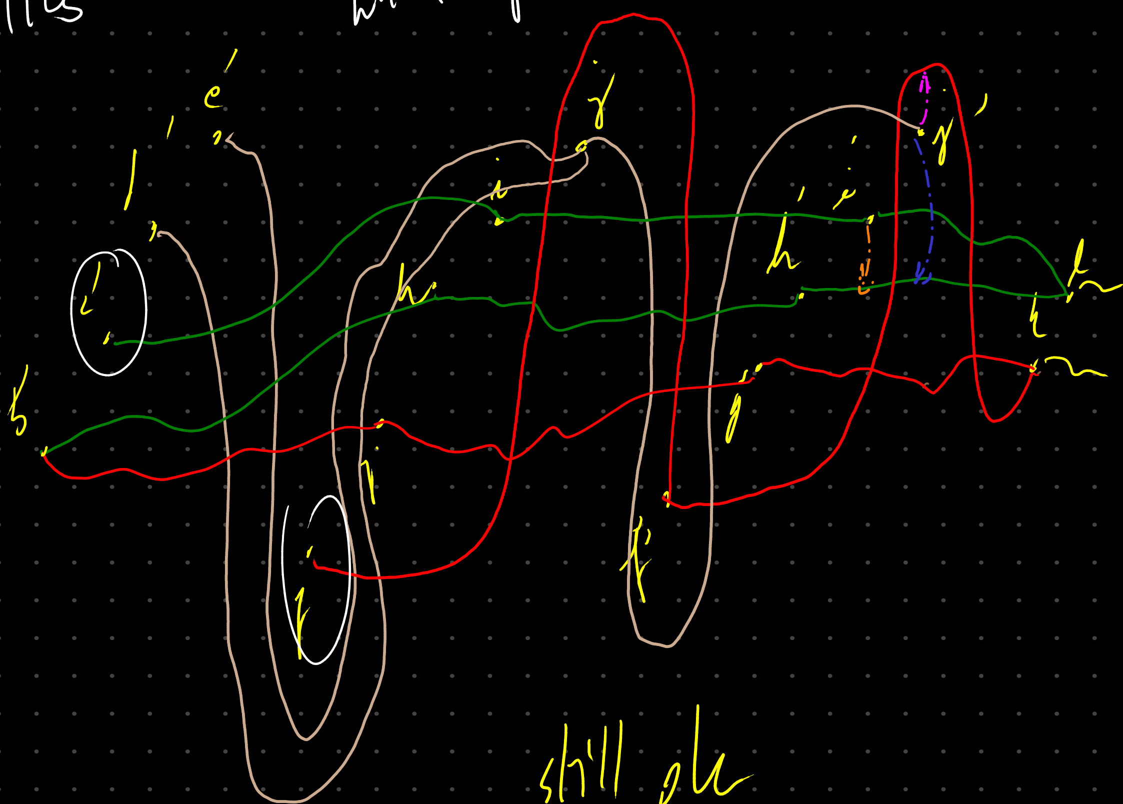




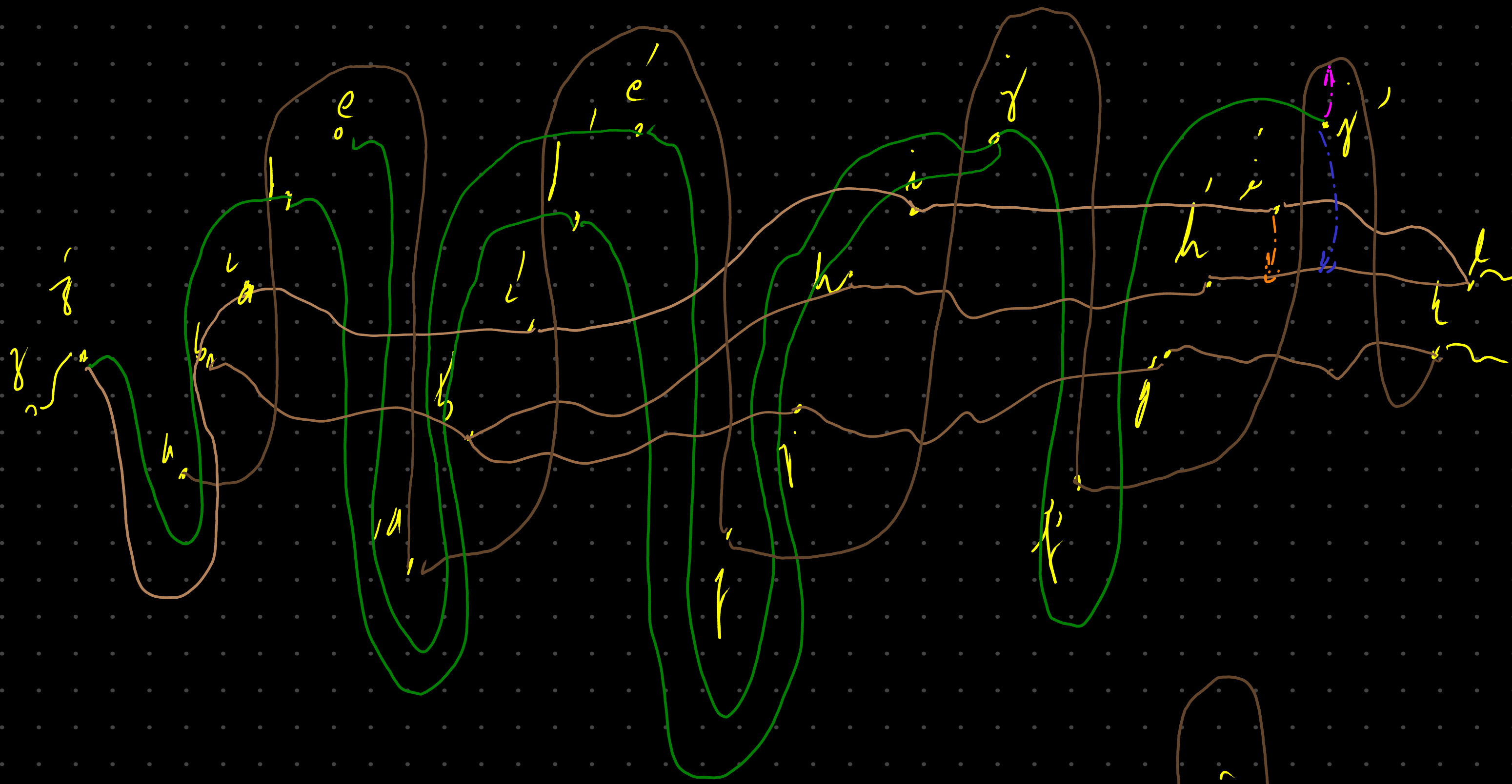


what if b was first

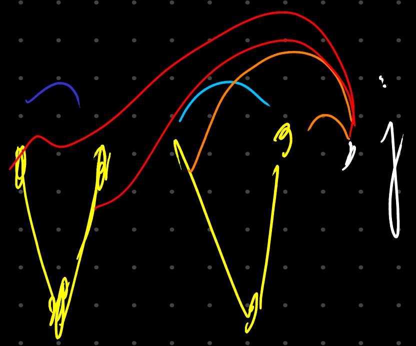
Nr of the vertices
in the same
level
are connected

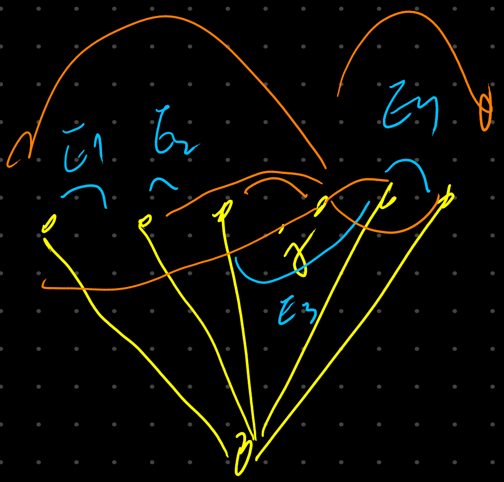


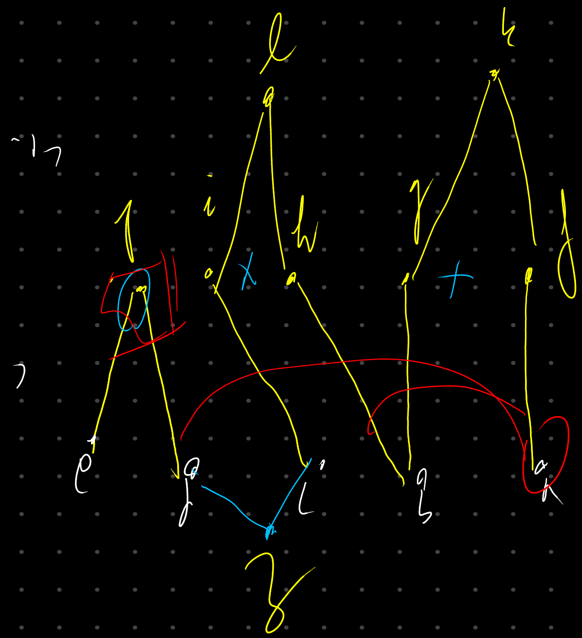
still ok

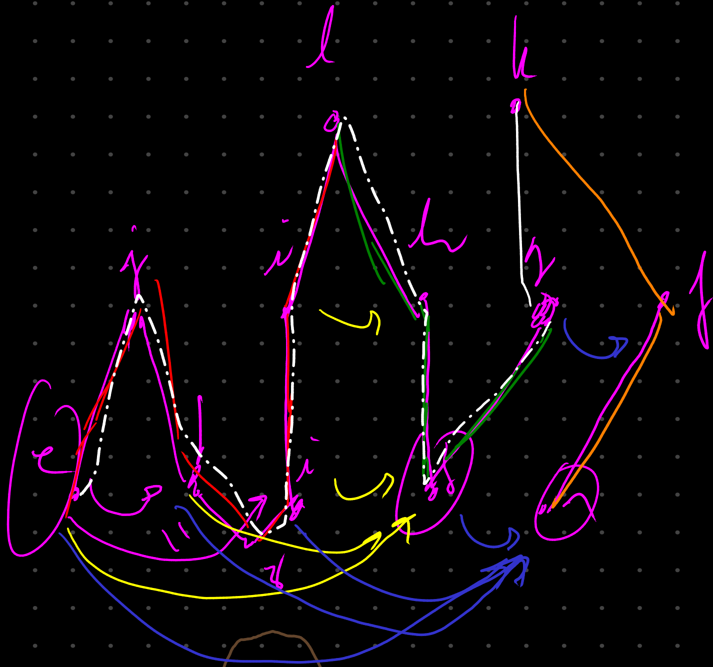


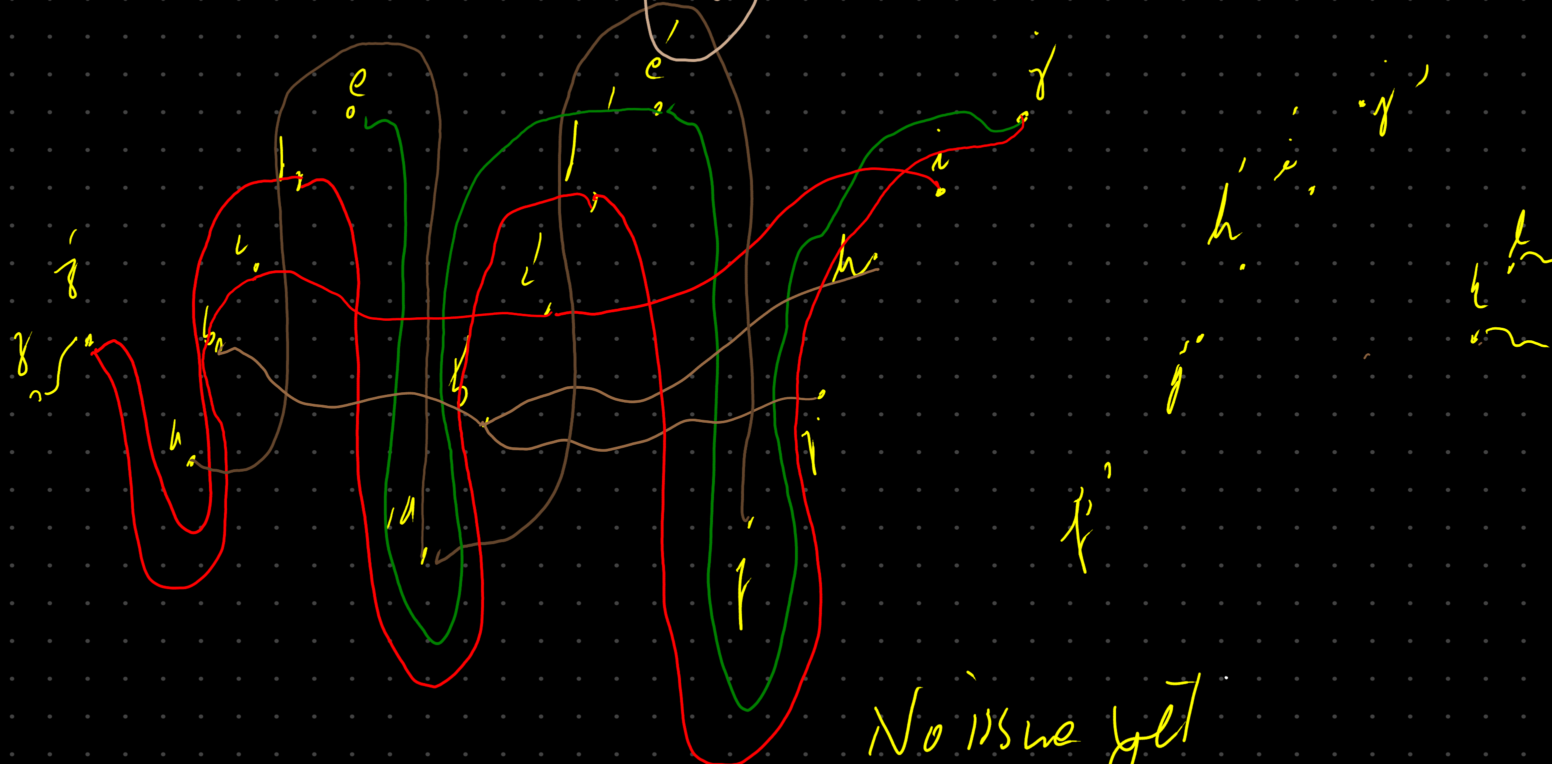
planarifikation;



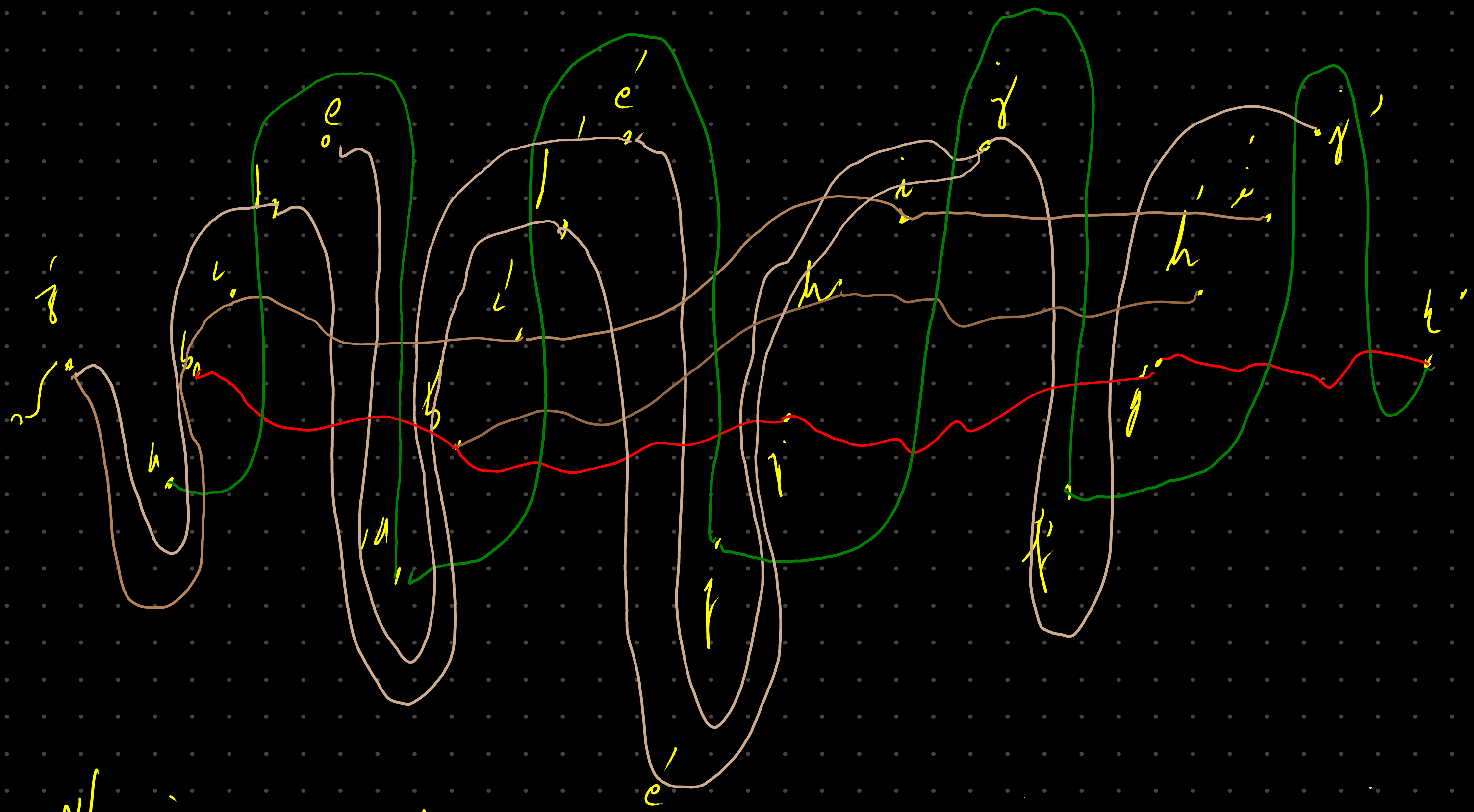




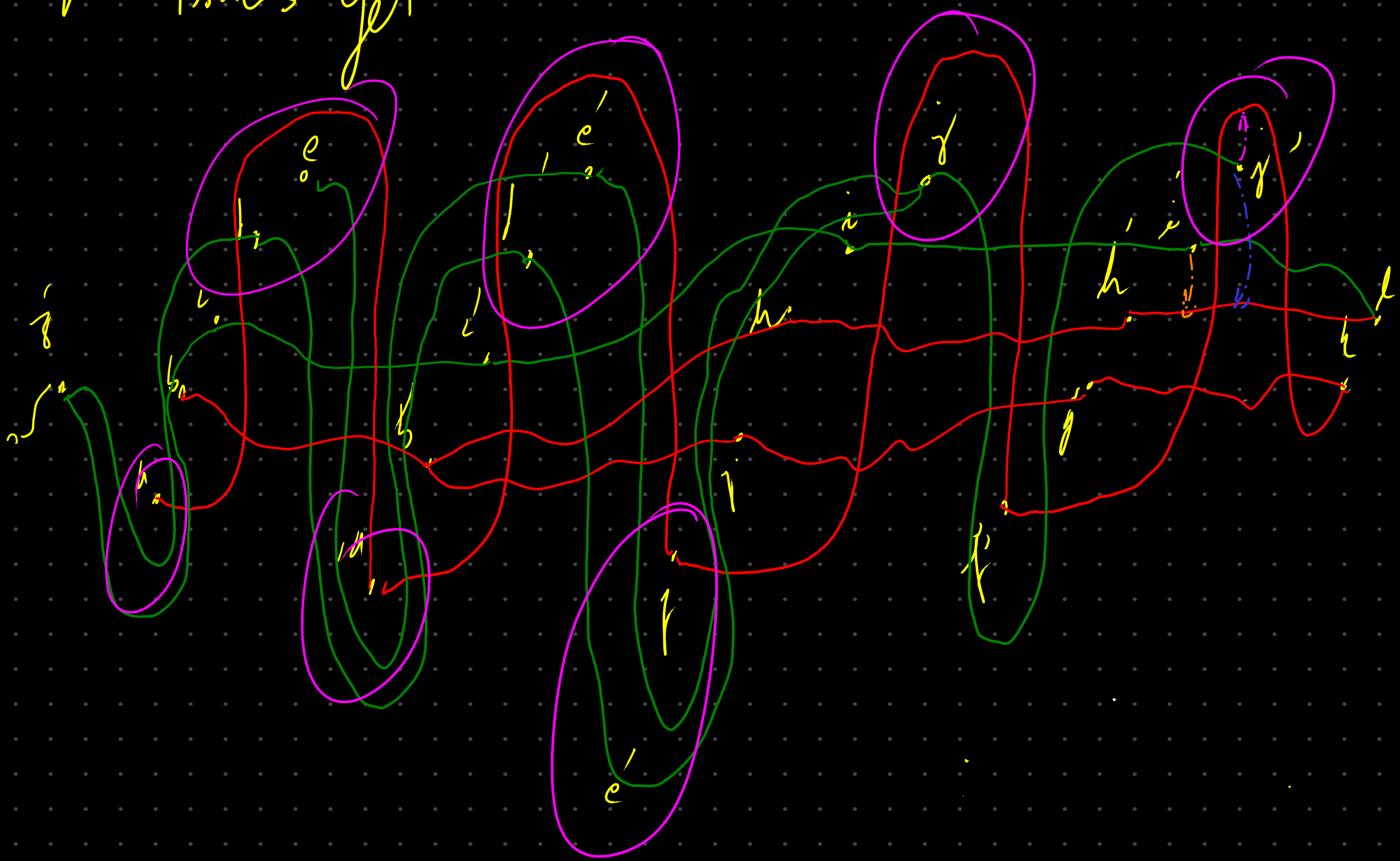




No issue yet



No issues yet



it's the same build

a is wanted

(c,d) to potted

E_{green}

$\varphi(a,c) = \varphi(a,d)$

$E_{orange} = E_{purple}$

b is wanted

$\varphi(b,c) = \varphi(b,d)$

$E_{orange} = E_{blue}$

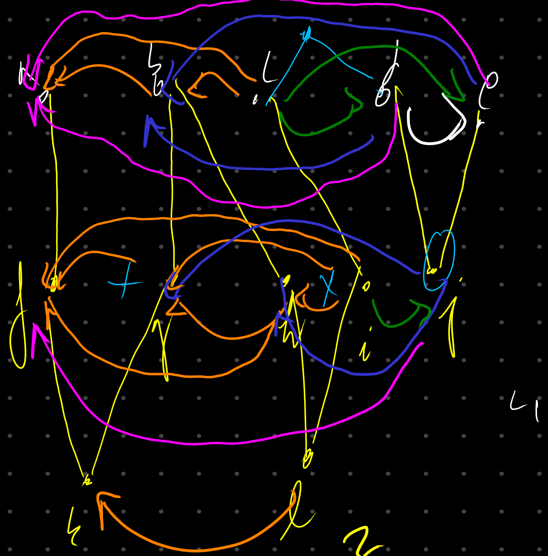
$E_o = E_{purple} = E_{blue} \;\Big|\; = 0\ 0\ 1\ 1$

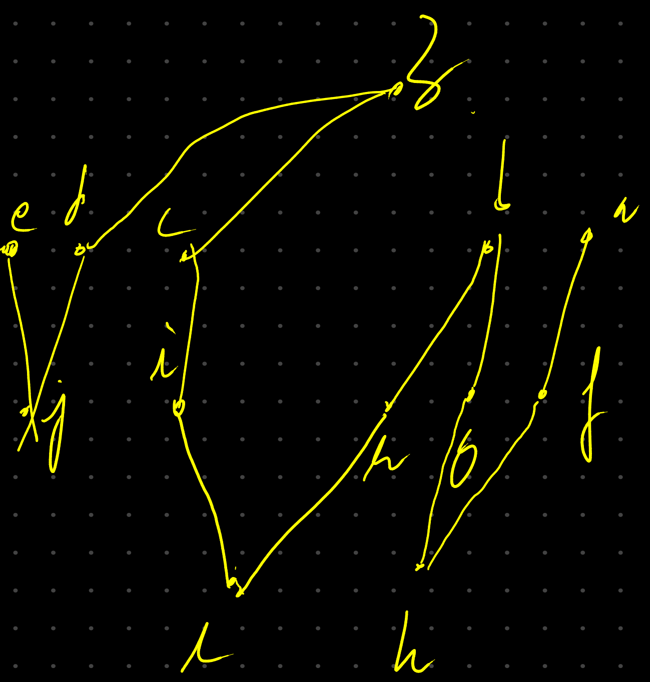
$E_{green} = E_w \;\Big|\; = 0\ 1\ 0\ 1$

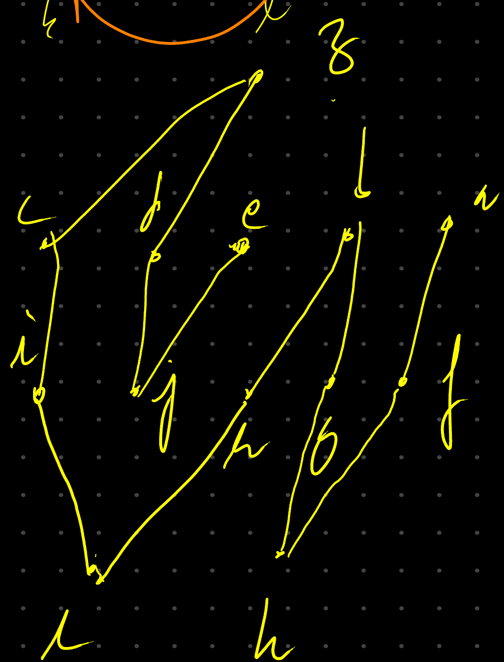
e is wanted

$\varphi(e,c) = \varphi(e,d)$

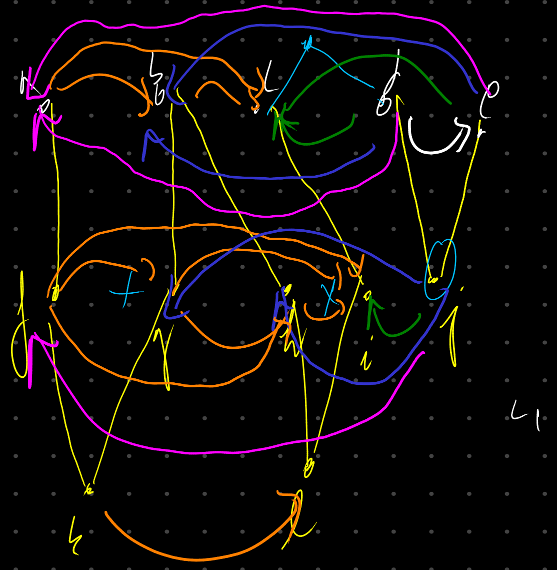
$\neg E_{green} = \neg E_{white}$







What if we start from the other side



j is wanted (f,g) and (h,i) to protect

$$\varphi(f,j) = \varphi(g,j)$$

$$E_{purple} = E_{blue}$$

$$\varphi(h,j) = \varphi(i,j)$$

$$E_{blue} = E_{green}$$

	0	1
$E_{purple} = E_{blue} = E_{green}$	0	1
E_{orange}	X	X
E_{white}	X	X